THINGS TO KNOW:

1. **Lab report must contain following sections: (order must be maintained)**
   a) **Title /Question**
   b) **Theory: The brief overview of the concept /techniques/syntax/technology used in the program**
   c) **Code: The complete code (markups, styles and scripts)**
   d) **Output: Screenshot of the output (in console and/or browser window)**
2. **Output screen should be captured (use snipping tool), printed and attached in the report. Other contents must be handwritten.**
3. **Every source code must include the printing statements to print following information after your main output:**

   Lab No.:

   Name:

   Roll No./Section :

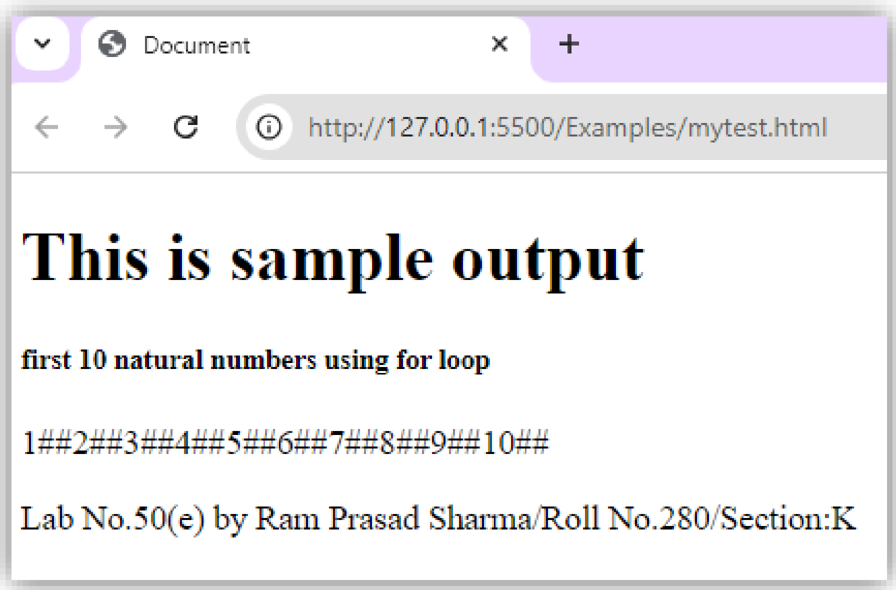
4. **Contents should be written on single side of A4 sized paper.**
5. **The works must be submitted within specified deadline.**
6. **Cover page and index page should be attached in the report appropriately.**

<u>Index Page Format (can be printed)</u>

List of Lab Works

Lab No.	**Title /Question**	**Submission Date**	**Signature**	**Remarks**
1(a)	This is sample	2079/11/12		
1(b)	This is also sample	2078/11/12		

Sample Output*: (must be readable, showing browser address bar or console window menus)*



[Lab reports after deadline will not be accepted. If you have any genuine reasons for not doing tasks in time inform the instructor before deadline!]

<u>Scripting Language / BCA – 4th Semester Lab Works (Part-2)</u>

12. Write a JS program to highlight all of the words over eight characters long in the paragraph text (with yellow background) when a button below the paragraph is clicked.
13. Create a webpage with following functionality.
    - ☞ There should be three radio buttons labeled with three colors. When one is selected the background of the page must be changed to respective color.
    - ☞ There should be an image, when mouse cursor is placed over the image, description of the image should be displayed on a paragraph below that image and when muse cursor is removed from the image, the description should be vanished.
14. Working with HTML DOM
    a) Create a HTML page and write a JS program to demonstrate the use of following properties of any node objects in DOM

`nodeName`	`nodeType`	`lastChild`
`nodeValue`	`parentNode`	`nextSibling`

`previosSibling` `childNodes` `firstChild`

b) Create a web page and write Script with JS for Selecting Document Elements as mentioned below:

- Selecting element by ID
- Selecting elements by name
- Selecting elements by tag type
- Selecting elements by CSS class
- Selecting element by querySelector()
- Selecting element by querySelectorAll()

c) Create a web page and write Script using JS for demonstrating

- Node creation
- Node insertion (using appenChild() and insertBefore())
- Node replacement (using replaceChild())
- Node Removal (using removeChild())

d) Demonstrate the use of following methods in DOM to work with attributes of HTML document

`getAttribute()` `setAttribute()`

`hasAttribute()` `removeAttribute()`

e) Create a form to enter the record of books. The interface should be as in following figure and descriptions:

Book id initially should be 100 and increments by 1 on every click of plus button. This value should be shown on textbox but cannot be editable by user. When plus button is clicked similar row should appear below it.

f) Create a JS array containing some districts of your provenience. Now display those districts in unordered list dynamically on web page using DOM manipulation.

15. Understanding Regular Expression in JS

a) Write a JS program to demonstrate the use of following concepts
   - two ways of creating regular expression pattern in JS
   - using **exec()** , **test(), match(), matchAll(), search(), replace(), replaceAll()** and **split()** functions.

b) Create and test for following Regex patterns in JS
   i. The first character of a string must be uppercase.
   ii. A given value must contain a question marks, a dash and an underscore.
   iii. The value must start from A or a and must end with full stop (.) or exclamation (!).
   iv. The value must contain only one blank space.
   v. The string must not contain any digits.
   vi. The string must conation at least a symbol from the group of symbol (, , @, $, _, %, !, &, *, ^, ~, -, +, (,), {,}, [,], . , :)
   vii. The value must not contain any special character and digits.
   viii. The value must contain @ symbol between two words each of at least one character.
   ix. The string must not start with digit.
   x. The string must start with underscore character or alphabets.
   xi. The value must start with 'BCA-TU-' and after these character there must be only digits.
   xii. The value must be in email address format.
   xiii. The value must contain at least a vowel letter.
   xiv. The value contain ten digits and first digit cannot be 0.
   xv. The value should be email address and its must end with 'edu.np'
   xvi. The value should be email address and it must contain 'gmail' or 'yahoo' after @ symbol.
   xvii. The regex must match if the string contains 'nepal' case insensitively.

16. Forms and Client Side Validation using JS.

a) Write a JS program to demonstrate the use of forms object (document.forms)

b) Create a registration form and validate using separate JS file (should be from scratch. API /Library is not allowed). The form must include following entries and validation rules:

- First Name (required , must contain alphabets only the length should be 1 to 50)
- Last Name (optional, must contain alphabets only the length should be 1 to 50)
- Date of birth (required, use date type)
- Gender (required, must be selected from combo box)
- Provenience (required, must be selected from combo box, 7 Provenience of Nepal)
- District (required, must be selected from combo box, 77 districts of Nepal)
- Cell Phone (required, 10 digits only not other symbols and alphabets)
- Email address (optional, if entered must be in email address format)
- User Name (required, must start with alphabet, can contain digits and only two special character (@ and _) are allowed, length must be in between 6 to 15 characters)
- Password (required, must contain at least a digit, a special character, an uppercase letter and a lowercase letter, length must be in between 8 to 20 characters)
- Confirm password (required, must match the value entered in password field)
- Register (a submit button)

Show the error message with guideline next to the element in red color if the value is not satisfying the rule.

17. Create a web application using JS to simulate simple calculator which can perform addition, subtraction, multiplication and division for both integer and floating point numbers. Your application should contain all the required digits and symbols, displaying area and necessary other designs. The application should be able to response the input from keyboard press and mouse click both. [you can print the whole content for this lab task]
18. Cookies in JS

Write a JS program to demonstrate following concepts:

- Different ways of cookie creation
- Reading cookie
- Changing cookie
- Deleting cookie

19. Some basic PHP programs
    a) Create a html form to take two numbers form text fields and display the multiplication table of bigger number using PHP.
    b) Create a html form to take two numbers form text fields and an operator (among +, -, *, /, %) from combo box and display the result on a label when user press a submit button by using switch-case-default syntax in PHP.
    c) Write a PHP program to demonstrate different ways to create arrays and display them using var_dump() and print_r() functions.
    d) Write a PHP program containing associative array and illustrate the use of for loop and foreach loop in accessing arrays.
    e) Write a PHP program to demonstrate following array functions

       array_push() , array_shift(), array_search(), array_unshift()

       sort(), asort(),ksort(), rsort(), arsort(), krsort()

       each(), current(), reset(), end(), next(), pos(), and prev()

       shuffle(), array_reverse()

20. Working with functions in PHP
    a. Write a PHP program having a function which take age as parameter and returns "You are restricted" if age is less than 18 otherwise returns "Welcome".
    b. Write a PHP program having a function that takes an array of number as parameter and prints the sum of second largest and smallest elements.
    c. Write a PHP program to demonstrate the function with default arguments.
    d. Write a PHP program to differentiate pass by value and pass by reference in PHP functions.
21. Write a program to demonstrate regular expression in PHP using the functions like preg_match(), preg_match_all() and preg_replace() .

22. Create a form with validation using PHP. The validation errors should be displayed after each field if invalid. Consider following validation rules.
    - All fields are required. (name,username,age,password and confirm password)
    - Name cannot have numbers and special characters
    - Username should be of minimum 8 character and must begin with alphabet
    - Age cannot be less than 16 years
    - Password and confirm password must match.
23. Working with date and time in PHP.
    a. Write a program to demonstrate the meanings of different formatting options available in date() and time() functions in PHP.
    b. Write a program to illustrate mktime() and strtotime() functions in PHP.
24. Handling files in PHP
    a. Write a program to read and write file using fgetc() and fputc() functions.
    b. Write a program to read and write file using fread() and fwrite() functions
    c. Write a program to copy the contents of one image file to another. (file copy)
    d. Write a program in PHP which reads a text file and write the contents into another file by replacing the words starting with "T or t" by "HEHE".
    e. WAP in PHP to demonstrate random accessing of file and associated functions.
    f. Create form to take values needed for a bio-data and generate a bio-data in pdf format using PHP.
25. Including and uploading files
    a. Write programs to demonstrate the use of include (), require(), include_once() and require_once() functions.
    b. Write a program to demonstrate the file uploading process in PHP.